

SVENSKA HOGARNA, Sweden

WMO: 024960

Lat: 59.444N

Lon: 19.506E

Elev: 12

StdP: 101.18

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-8.3	-6.7	-12.6	1.3	-7.0	-11.0	1.5	-5.8	19.5	-0.5	17.8	-0.3	8.2	50	0.840

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.6	23.3	19.4	21.9	18.6	20.6	17.6	20.2	22.6	19.0	21.2	18.0	20.1	4.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.0	21.6	18.0	13.0	20.5	17.1	12.2	19.6	58.0	22.8	53.9	21.2	50.9	20.2	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.1	15.3	13.8	DB	-9.7	24.8	2.9	2.1	-11.8	26.2	-13.5	27.5	-15.2	28.6	-17.3	30.1
			WB	-10.5	20.4	2.7	1.7	-12.5	21.7	-14.0	22.7	-15.6	23.6	-17.5	24.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-0.2	-1.0	0.6	3.6	7.8	12.8	17.2	17.5	13.5	8.5	4.5	1.8
	DBStd	6.91	2.91	3.29	2.49	1.91	2.35	2.32	2.60	2.34	2.35	2.60	2.61	3.03
	HDD10.0	1666	315	308	291	192	76	3	0	0	2	60	164	255
	HDD18.3	4076	573	541	549	442	327	167	53	44	146	306	414	513
	CDD10.0	671	0	0	0	0	7	87	224	233	107	13	0	0
	CDD18.3	38	0	0	0	0	0	0	19	18	1	0	0	0
	CDH23.3	33	0	0	0	0	0	0	20	13	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	7.3	8.7	8.1	7.4	6.6	6.1	5.9	5.7	6.1	7.0	8.0	8.6	9.0
(15) Precipitation	PrecAvg	453	35	24	22	25	25	33	48	54	52	48	47	41
	PrecMax	672	80	55	57	64	71	81	146	125	123	140	94	99
	PrecMin	323	2	2	1	1	5	2	7	8	5	12	11	9
	PrecStd	75	18	12	12	14	17	19	32	30	30	26	21	23
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.7	5.0	6.7	10.1	15.8	20.4	25.2	24.8	20.0	14.7	10.1	8.0
		MCWB	4.8	3.4	4.3	6.8	12.1	15.7	19.8	20.1	16.5	13.0	9.0	7.0
	2%	DB	4.9	4.0	5.3	8.4	13.5	18.5	23.6	23.3	18.6	13.6	9.4	7.1
		MCWB	4.2	2.9	3.5	5.9	10.4	14.7	19.7	19.3	15.7	12.1	8.4	6.3
	5%	DB	4.3	3.5	4.4	7.2	12.2	17.2	22.3	22.1	17.4	12.6	8.6	6.3
		MCWB	3.6	2.6	2.9	5.1	9.6	14.1	18.9	18.6	15.0	11.2	7.7	5.4
	10%	DB	3.6	2.9	3.6	6.3	11.1	16.0	21.1	20.8	16.5	11.7	7.9	5.5
		MCWB	2.8	2.1	2.4	4.5	8.8	13.4	18.1	17.7	14.4	10.4	7.0	4.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.2	3.9	4.8	7.1	12.7	16.6	21.6	21.7	17.4	13.6	9.6	7.2
		MCDB	5.5	4.3	6.1	9.3	15.3	19.1	23.4	23.7	19.0	14.2	9.8	7.7
	2%	WB	4.4	3.2	3.8	6.1	10.9	15.3	20.6	20.1	16.2	12.4	8.6	6.5
		MCDB	4.8	3.7	4.8	8.0	12.9	17.5	22.8	22.5	17.9	13.2	9.1	7.0
	5%	WB	3.7	2.7	3.2	5.4	10.0	14.5	19.4	19.0	15.5	11.6	7.9	5.6
		MCDB	4.2	3.2	4.1	6.9	11.7	16.6	21.8	21.2	16.9	12.4	8.5	6.1
	10%	WB	2.9	2.2	2.5	4.7	9.1	13.7	18.3	18.2	14.7	10.7	7.2	4.8
		MCDB	3.6	2.8	3.3	6.0	10.6	15.7	20.6	20.3	16.0	11.5	7.8	5.3
(35) Mean Daily Temperature Range	5% DB	MDBR	2.6	2.6	2.9	3.5	4.0	4.1	3.9	3.6	3.0	2.4	2.3	2.5
		MCDBR	2.6	2.8	3.8	4.9	5.7	5.6	5.2	4.8	3.8	2.8	2.6	2.5
		MCWBR	2.8	2.4	2.8	3.1	3.4	3.0	2.8	2.6	2.5	2.5	2.8	2.7
	5% WB	MCDBR	2.6	2.5	3.5	4.6	5.2	5.2	4.7	4.4	3.4	2.6	2.3	2.4
		MCWBR	2.8	2.4	2.7	3.1	3.4	3.1	2.8	2.7	2.5	2.6	2.6	2.6
(40) Clear-Sky Solar Irradiance	taub		0.287	0.321	0.338	0.375	0.370	0.362	0.391	0.381	0.351	0.338	0.312	0.272
	taud		2.223	2.297	2.349	2.334	2.426	2.489	2.429	2.442	2.509	2.412	2.238	2.134
	Ebn at Noon		507	686	797	826	860	872	834	814	784	664	461	371
	Edh at Noon		43	68	87	105	103	98	102	94	75	61	43	32
(44) All-Sky Solar Radiation	RadAvg		0.27	0.80	2.31	4.04	5.44	6.03	5.63	4.43	2.81	1.21	0.37	0.18
	RadStd		0.04	0.15	0.23	0.37	0.44	0.33	0.51	0.36	0.29	0.19	0.09	0.02

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	-121	-159	N/A	N/A
(47) Regional (48 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-110	-131	N/A	N/A

Nomenclature: See separate page